MAT2040

Tutorial 8

CUHK(SZ)

July 1, 2024

Question 1

Compute the determinant of following matrices

(a) $A = \begin{bmatrix} 5 & 0 & 0 \\ 4 & 4 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

(b) $B = \begin{bmatrix} 2 & -3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \end{bmatrix}$

(c) $C = \begin{bmatrix} 3 & -3 & 9 \\ 2 & -5 & 7 \\ 3 & -3 & 9 \end{bmatrix}$

Question 1 - Solution

$$\text{(a) } \det(A) = \begin{vmatrix} 5 & 0 & 0 \\ 4 & 4 & 0 \\ 0 & 1 & 2 \end{vmatrix} = 5 \begin{vmatrix} 4 & 0 \\ 1 & 2 \end{vmatrix} = 5 \times 4 \times 2 = 40.$$

$$\text{(b) } \det(B) = \begin{vmatrix} 2 & -3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \end{vmatrix} = 2 \begin{vmatrix} 1 & -1 \\ 0 & -5 \end{vmatrix} = 2 \times 1 \times (-5) = -10.$$

$$\text{(c) } \det(C) = \begin{vmatrix} 3 & -3 & 9 \\ 2 & -5 & 7 \\ 3 & -3 & 9 \end{vmatrix} = 0 \text{ as } C \text{ has two identical rows.}$$

Question 2

Let $a, b, c, d \in \mathbb{R}$ and

$$\mathbf{A} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix}$$

(a) Show that

$$\det(\mathbf{A}) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$$

(b) Show a, b, c, d are distinct if and only if A is nonsingular.

Question 2 - Solution

(a) Operate $-aR_3 + R_4, -aR_2 + R_3, -aR_1 + R_2$ successively and expand along the first column, we get

$$\det(\mathbf{A}) = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b(b-a) & c(c-a) & d(d-a) \\ 0 & b^2(b-a) & c^2(c-a) & d^2(d-a) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{vmatrix}$$

Question 2 - Solution

Operate $-bR_2 + R_3, -bR_1 + R_2$ and expand along the first column, we get

$$\begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & c-b & d-b \\ 0 & c(c-b) & d(d-b) \end{vmatrix}$$

$$= (c-b)(d-b)\begin{vmatrix} 1 & 1 \\ c & d \end{vmatrix}$$

$$= (c-b)(d-b)(d-c)$$

Therefore,

$$\det(\mathbf{A}) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$$

Question 2 - Solution

(b) a, b, c, d are pairwise distinct $\leftrightarrow$ none of the terms $d-a, c-a, b-a, d-b, c-b, d-c$ is 0 .

$$\leftrightarrow (d-a)(c-a)(b-a)(d-b)(c-b)(d-c) \neq 0$$

$$\leftrightarrow \det(\mathbf{A}) \neq 0$$

$$\leftrightarrow \mathbf{A} \text{ is nonsingular.}$$

$\leftrightarrow \mathbf{A}$ is nonsingular.

Question 3

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = 0$$

Calculate the value of x

Solution

Here sum of entries of each column is same $(= 5x + 4)$,
so let us operate $R_1 \to R_1 + R_2 + R_3$ to make all entries of first row equal $(= 5x + 4)$.

$$\text{L.H.S.} = \begin{vmatrix} 5x+4 & 5x+4 & 5x+4 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$$\xrightarrow{\text{take common from } R_1} (5x+4) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$$\xrightarrow[C_2 \to C_2 - C_1]{C_3 \to C_3 - C_1} \begin{vmatrix} 1 & 0 & 0 \\ 2x & 4-x & 0 \\ 2x & 0 & 4-x \end{vmatrix}$$

$$= (5x+4)(4-x)^2 = 0$$

Thus $x = -\frac{4}{5}$ or $x = 4$

Question 4

Compute the determinant of following matrix

$$A = \begin{bmatrix} 3 & 2 & 0 & 4 \\ 2 & 4 & 0 & 0 \\ 3 & 6 & 2 & 20 \\ 4 & 8 & 0 & 1 \end{bmatrix}$$

Solution

Expanding along the third column yields,

$$\begin{aligned}
\det(A) &= (2)(-1)^{(3+3)} \begin{vmatrix} 3 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 8 & 1 \end{vmatrix} \\
&= 2\left((4)(-1)^{(1+3)} \begin{vmatrix} 2 & 4 \\ 4 & 8 \end{vmatrix} + (1)(-1)^{(3+3)} \begin{vmatrix} 3 & 2 \\ 2 & 4 \end{vmatrix} \right) \\
&= 2(4 \times 0 + 4 \times 8) \\
&= 16.
\end{aligned}$$